

A PRFT score definition

Workflow used to define the transcriptomic PRFT score carried through the manuscript.

Proteostasis score

Bulk transcriptomic axis for protein-folding and stress-adaptation genes.



Ferroptosis-tolerance score

Transcriptomic axis for ferroptosis-defense and tolerance-related genes.



PRFT score

Proteostasis score plus ferroptosis-tolerance score; median split used in TCGA-LAML.



Downstream use

Risk-group assignment used for discovery, survival modeling, and external validation.

B Candidate derivation workflow

Discovery layers retained in workflow form only; detailed DEG and WGCNA visual outputs are kept in Supplementary Figure S1.

PRFT-high DEGs

Differential-expression screening between median-defined PRFT groups.



Intersect with PRFT-positive WGCNA modules

Shared genes between PRFT-high differential expression and PRFT-positive co-expression modules.



715 candidates

PRFT-associated candidate pool carried forward for cross-platform feasibility filtering.



33 cross-platform candidates

GPL570-detectable subset retained for survival-model screening and robustness checks.

Detailed DEG and WGCNA visual outputs are preserved in Supplementary Figure S1.

C Model-locking summary

Locked-model decision path retained for the final manuscript package.

Survival filtering

Candidate genes screened in TCGA-LAML and cross-platform validation-ready space.



Machine-learning robustness

Clinical Model 2 and survival-learning comparisons supported retention of the fixed model.



Locked six-gene Formula A

CLCN5, ARHGEF5, TRIM32, ITGB2, SAT1, and ACOX2 carried forward unchanged.



Coefficient display policy

Locked Formula A coefficients are shown in Figure 2A only. No alternative coefficient panel is displayed here.

D Downstream evidence map

Evidence streams carried into the bounded interpretation model.

TCGA / external validation

Figure 2 training and GPL570 validation support.



Clinical + ML robustness

Clinical adjustment and survival-learning comparisons retained the fixed model.



Bulk signature context

Stress-adapted, myeloid/immune-associated transcriptomic patterns.



Processed single-cell support

Processed-table support only; no de novo single-cell analysis.



BeatAML ex vivo associations

Pharmacogenomic associations interpreted as ex vivo hypotheses only.

This schematic summarizes manuscript logic and avoids duplicating discovery panels already retained in Supplementary Figure S1.